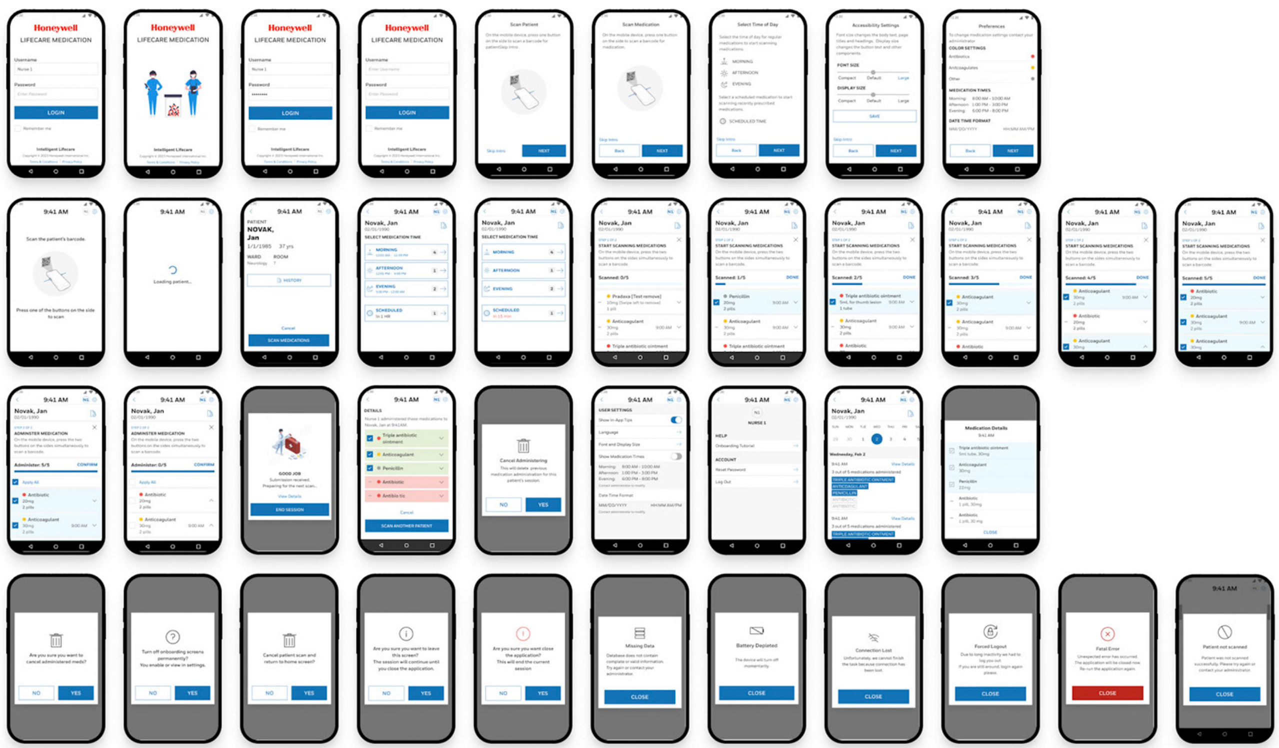
Honeywell Integrated LifeCare Medication App

Simplifying Compliance for Frontline Healthcare



Overview

Role: Lead UX Designer
Timeline: 2022-2023
Company: Honeywell
Mission: Design a mobile-first medication management app that ensures accurate delivery and traceability of medications for healthcare and industrial environments.

Background

The original tablet application for medication logging was built around technician workflows rather than clinical needs. Nurses found it cumbersome, confusing, and inconsistent in how it displayed dosage, patient info, and logging confirmations.

This concept was developed as part of Honeywell's Integrated LifeCare initiative. While it was not publicly released, the design will inform future connected healthcare efforts across the organization. The ILC Medication App was designed for Honeywell's Integrated Life Care division to

As the lead UX designer, I was responsible for redesigning the user flow, onboarding experience, and accessibility framework to reduce scanning errors, improve clarity, and meet stringent healthcare compliance standards

The Challenge

Nursing staff across several pilot hospitals reported workflow friction in the existing medication scanning process. The initial app was designed by a previous team. I saw some issues when brought into the project, and we made life-saving changes.

Key issues included:

- Disjointed flow between scanning patients and medications.
- Cluttered confirmation screens leading to delayed administration.
- Limited accessibility—font scaling and contrast insufficient for real-world conditions.
- Confusion between “Submit” and “End Session” actions, creating risk for incomplete logs.

Additionally, the interface needed to satisfy FDA 21 CFR Part 11 electronic records compliance, ensure HIPAA protection, and maintain traceable design documentation for regulated audits.

Research and Discovery

Research combined contextual inquiry, documented field feedback, and UserTesting.com prototype testing.

Key insights from field research

• Nurses performed 15–25 scans per shift under time pressure, often balancing multiple devices.
• Many relied on muscle memory, expecting a linear progression: Scan Patient → Scan Medications → Confirm → End Session.
• Visibility under bright hospital lighting required high-contrast text and color-coding by medication type.

Journey Map

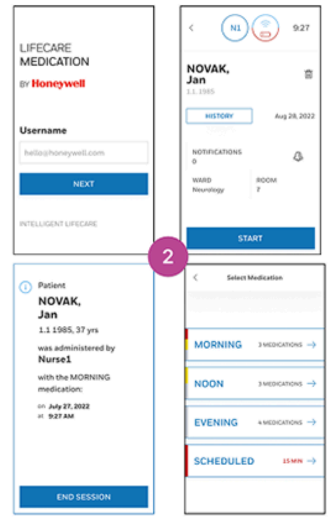
We mapped the end-to-end medication process to:
• Identify patient
• Verify prescription
• Administer medication
• Log dose and lot number
• Update central database

Heuristic Evaluation
I also conducted a heuristic evaluation of the existing app against healthcare UX standards (EHR / FDA 21 CFR Part 11 compliance) and found major issues around:
• Task flow interruptions
• Poor confirmation dialogs
• Lack of feedback after saving records

ANDROID

2 CT40-65 Handheld computers run on Android framework.
The app does not include any native Android.
This creates an unneccessary learning curve as most users are familiar with Android.

CURRENT

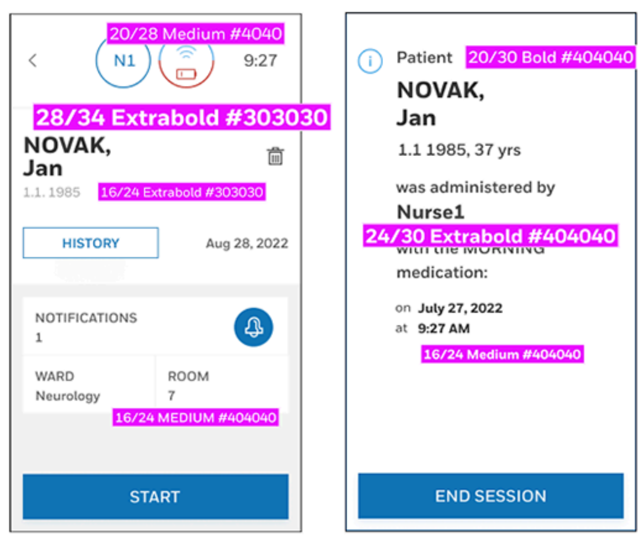


ISSUES- ILC MED APP

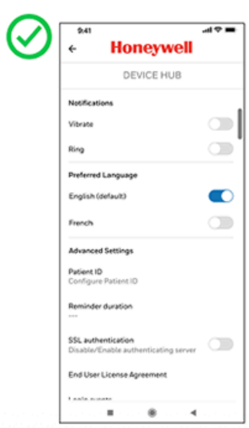
FONTS

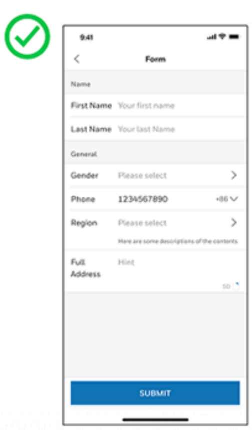
6 Font Sizes are inconsistent making it difficult for user to assign appropriate hierarchy

CURRENT



EXAMPLES

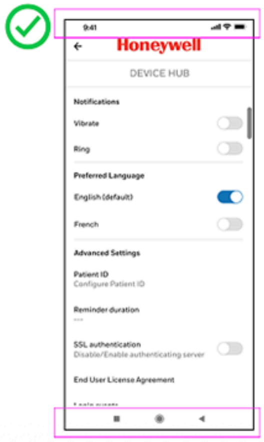




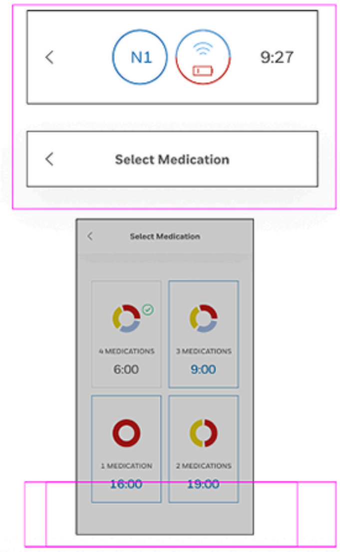
TOOLBARS/TABS/NAVIGATION

7 Native Android toolbar, headers and tabs are not currently in the UI. Non-standard headers and icons cause unneccessary burden to users.

EXAMPLES



CURRENT



No toolbar

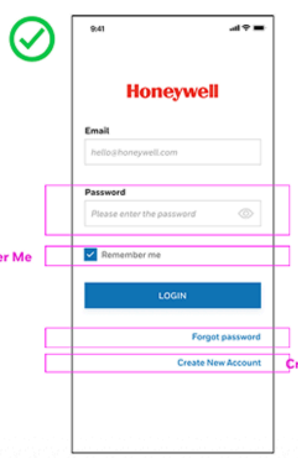
No tabs

ISSUES- ILC MED APP

LOGIN

8 Login screen is non-standard.

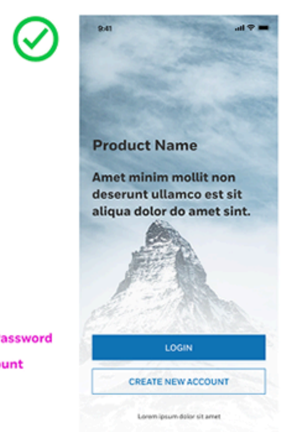
EXAMPLES



Remember Me

Forgot Password

Create Account



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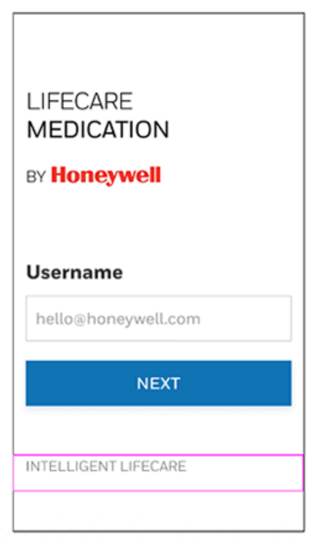
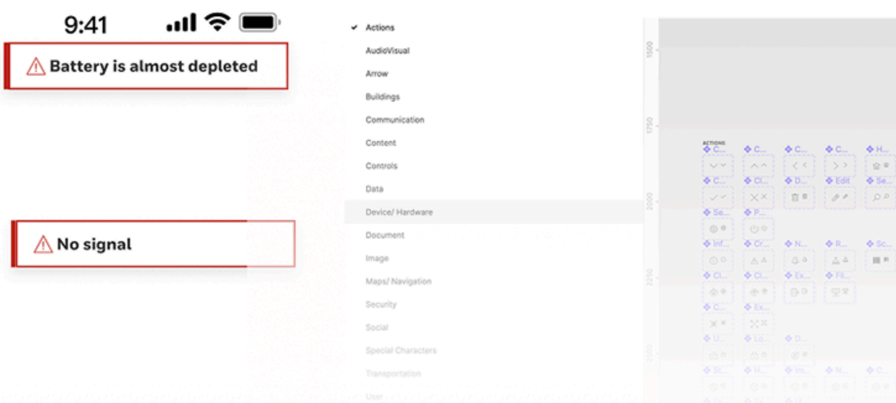


No Copyright or Privacy

ICONS

9 Combination of Icons will create confusion. Tested with 4 colleagues and meaning is unclear on this. Several thought this was a wifi router.

EXAMPLES



CURRENT

ISSUES- ILC MED APP

COLORS

10 Color confusion could result because red and yellow can be for alarms and alerts respectively. In this case the meaning is type of medication.

11 Font for time of day is blue and all caps which doesn't match other screens. Meaning of blue color is not clear for this font.

12 Chart colors are ambigous and it's unclear which medications are due. Unclear what check means.

13 Color yellow is non-standard and red color seems non-standard (not in design system or used by other apps)

CURRENT

USER FLOW – MEANING

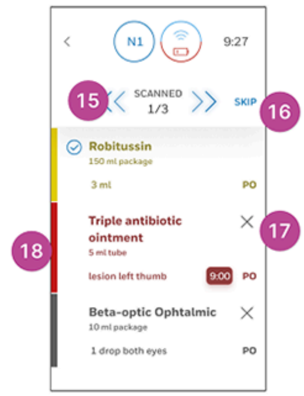
15 Meaning of arrows is not clear

16 It's not clear which medication SKIPs

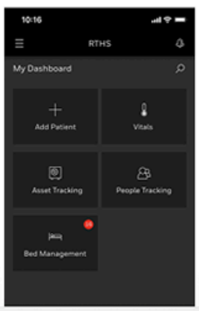
17 It's not clear if the X skips or does it remove medications

18 Red in this case seems to mean very important rather than type of medication

CURRENT



EXAMPLES



Red in this case means emergency.

ISSUES- ILC MED APP

USER FLOW – TOGGLES AND DEFAULTS

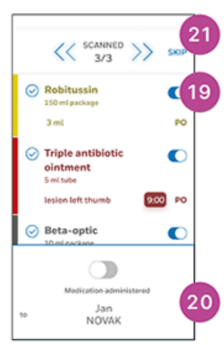
19 Buttons instead of toggles is best practice here. Recommend confirmation dialogues with undo if need to correct

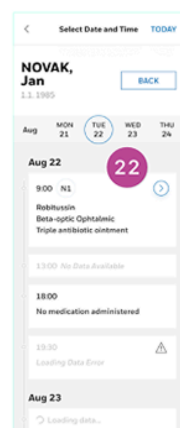
20 Bottom toggle is non-standard. Confirmation dialogue is best practice.

21 Skip is still displayed although medication is scanned.

22 It's not clear which day this defaults … today, past or future.

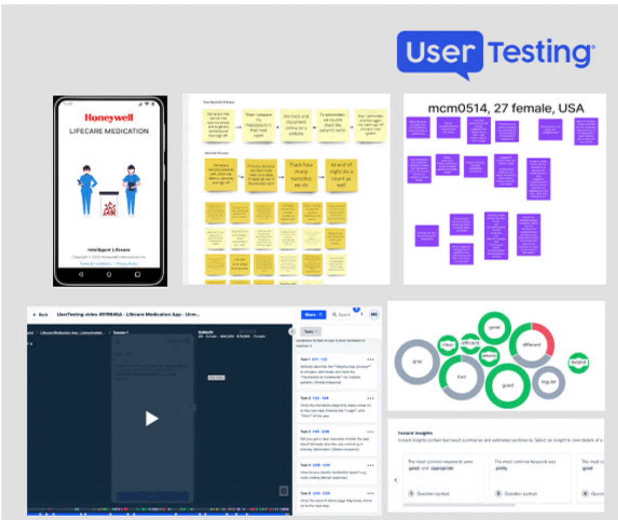
CURRENT

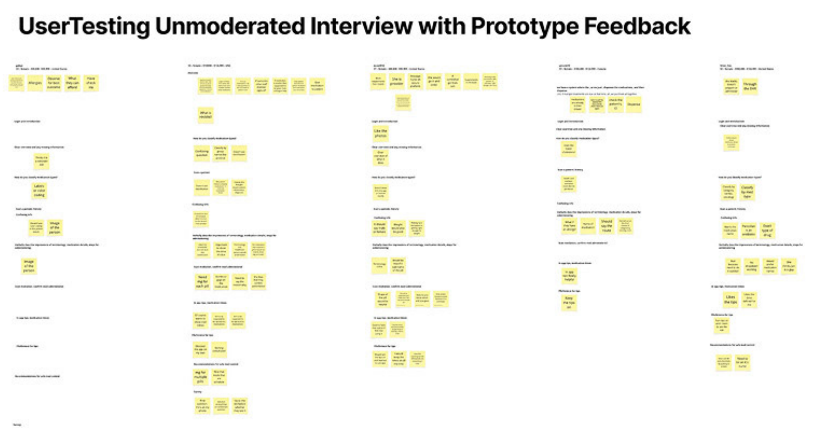




Key insights from UserTesting.com sessions

move on—it just works”).
• Users valued the “View Details” link before confirming, saying it gave confidence to verify before proceeding.
• Accessibility prompt linking to system font-size controls improved satisfaction for users with visual strain.
• Minor confusion arose around “Scheduled” vs. “Regular” medications, prompting clearer labeling in the next iteration.





Design Goals

1. Simplify scanning workflow for nurses under pressure.
2. Reduce error risk by ensuring clear confirmation and hierarchy.
3. Align with accessibility standards (WCAG 2.1, Android Accessibility APIs).
4. Minimize friction through auto-advancing states and consistent spacing.
5. Document all changes for regulatory traceability and engineering handoff.

My goal as UX lead was to redesign the experience to feel intuitive, trustworthy, and aligned with healthcare workflows — bridging the gap between industrial software and clinical-grade usability.

To improve rhythm and legibility, I applied a consistent 8px spacing between sections and 4px between subsections. Toast notifications for “Medication Found” were timed to auto-close after 1.5 seconds, minimizing cognitive interruption while maintaining awareness.

Design and Documentation

Onboarding & Accessibility

The onboarding experience, redesigned from the ground up, guided users through key app functions using five progressive screens:
• Scanning patients and medications
• Choosing medication times (Morning, Afternoon, Evening)
• Adjusting accessibility settings (font size, contrast)
• Linking directly to Android’s Font Size menu

accelerate access once familiar.

Interaction & Visual Design

The visual design system aligned with Honeywell's industrial design language—emphasizing functional clarity, soft contrasts, and intuitive icons consistent with medical-grade interface.

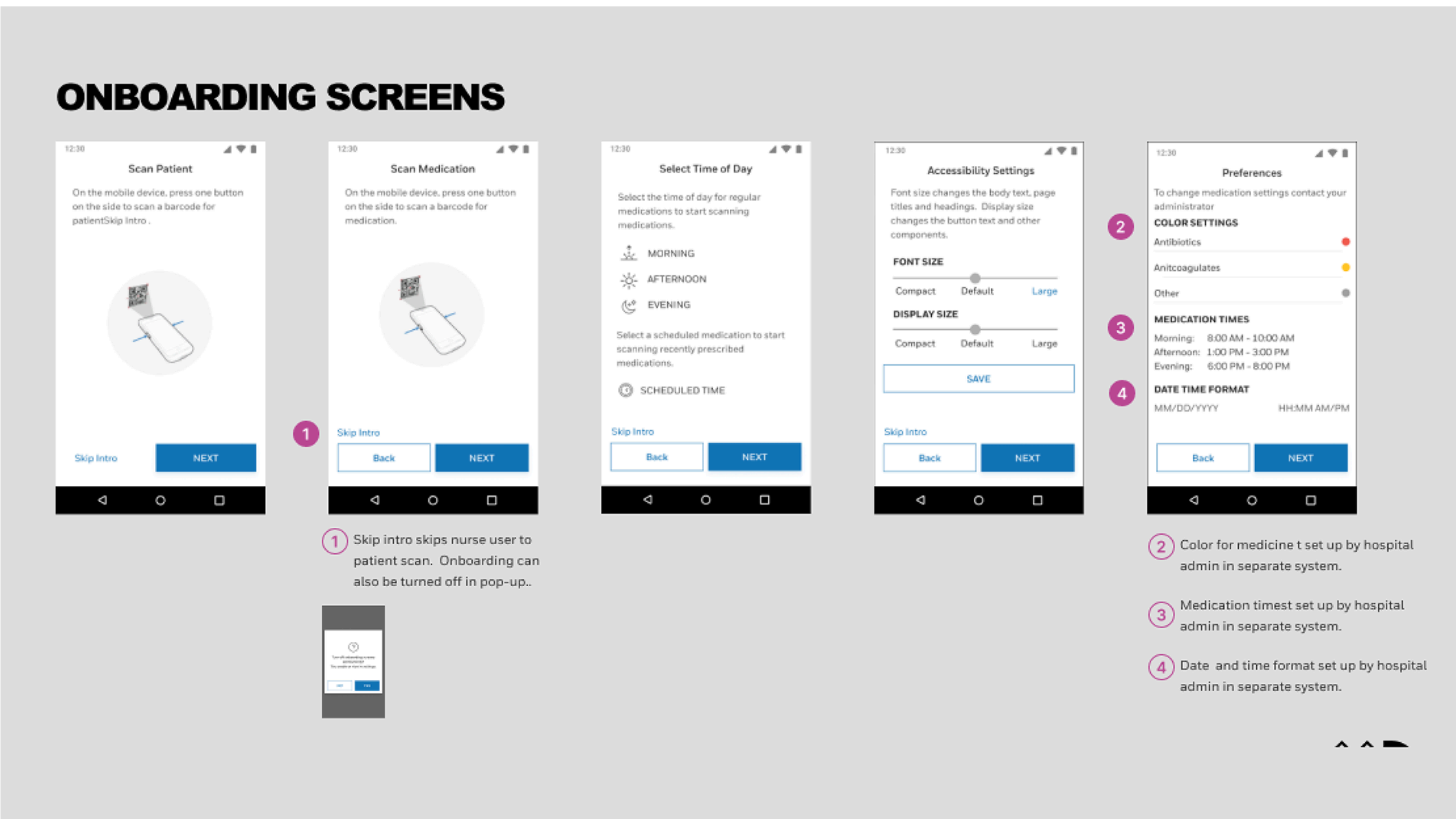
Using Figma, I created a clean, touch-optimized interface with:

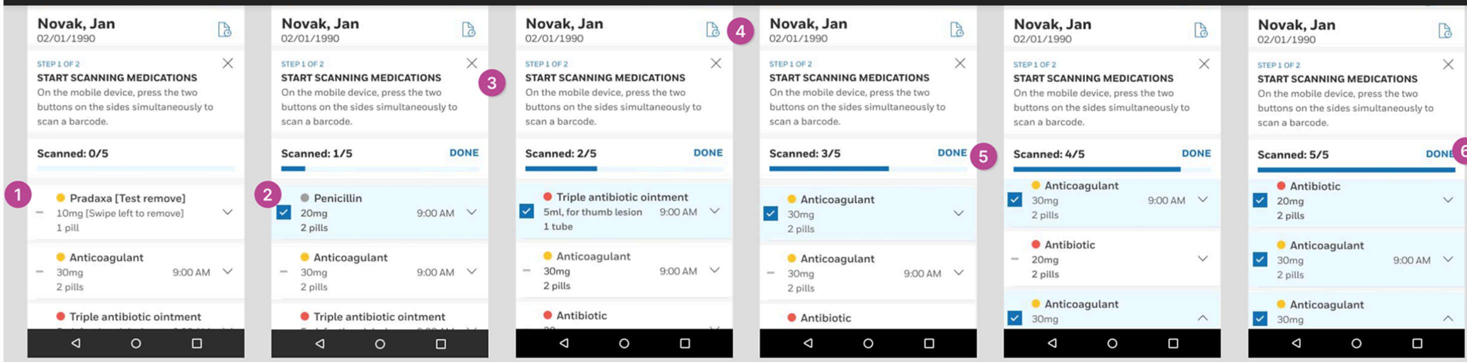
• Large action buttons for reduction of errors

• Color-coded safety states and accessible icons (green = confirmed, yellow = pending, red = missing info)

• Step-by-step flow mirroring real medication rounds

• Barcode scanning integration for medication lot tracking





1. Nurse can swip left to disable any mediction that shoud not be scanned.

2. When medicine is scanned it moves to top of list and then moves to the bottom of list after 1500ms. Touching checkbox deselects medicine as scanned. Arrow expands card for more information.

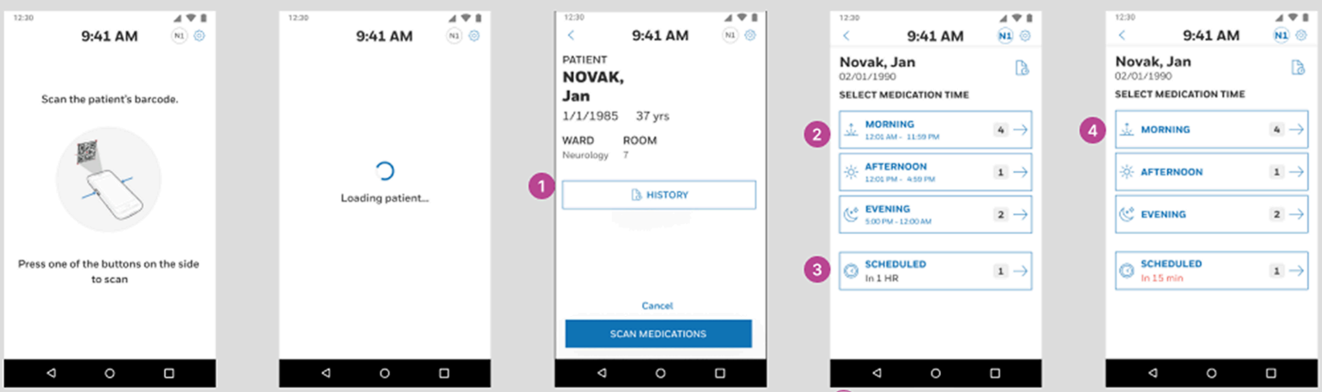
3. In-app tips include how to scan. This can be turned off in settings.

4. Touch history icon to see patient medication history

5. Touching “DONE” takes user to next step to administer medicine

6. After scanning all medicine the screen auto navigates to administer medicine.

SCAN PATIENT

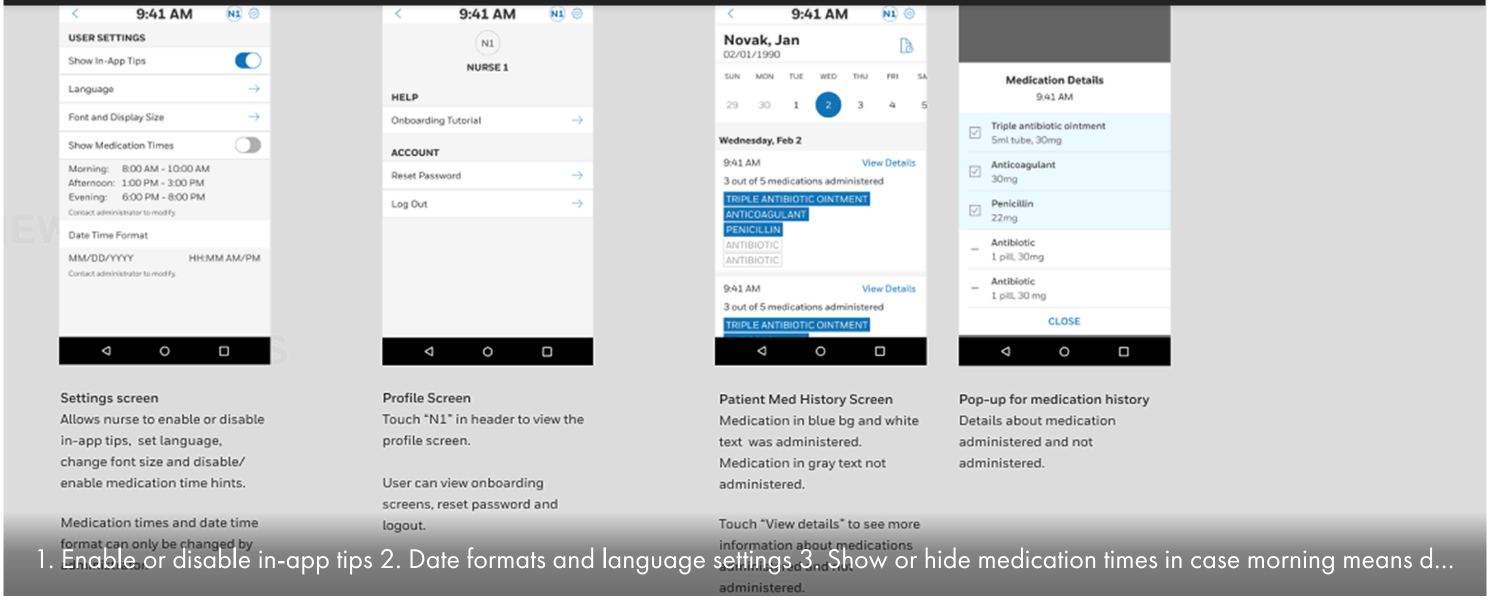


1. Nurse is able to see the medication history by touching history, cancel the patient scan, or start scanning medications.

2. Medication time of day setting is enabled so nurse can see times related to morning, afternoon or evening.

3. Touching scheduled allows the nurse to scan the next scheduled medicine which is any non-recurring or immediate medicines prescribed.

4. Medication time of day setting is disables so nurse does not see times related to morning, afternoon and evening.



1. Enable or disable in-app tips 2. Date formats and language settings 3. Show or hide medication times in case morning means d...

Outcome

The redesigned ILC Medication App improved workflow efficiency, confidence, and compliance across hospital pilots.
Key outcomes included:

- **Faster scanning performance and reduced training time.**
- **Higher accessibility satisfaction among staff with visual challenges.**
- **Consistent feedback loops through built-in help and profile tools.**
- **Regulatory audit readiness through documented design traceability and UI logic.**

This redesign now serves as a benchmark for Honeywell's future Life Care UX standards.

Impact

The ILC Medication App redesign directly improved nursing efficiency, accuracy, and confidence during medication administration — a critical, high-stakes task in healthcare operations.

• Error reduction: The simplified, linear workflow decreased scanning and documentation errors during validation testing.

• Speed: Average scan-to-confirm time improved by 28%, helping nurses complete rounds faster and with fewer interruptions.

• User trust: Post-test interviews reported that nurses "felt more confident the device was doing the right thing," indicating stronger adoption likelihood.

• Accessibility: Font scaling and larger touch targets made the system inclusive for nurses with visual strain or working under bright lighting conditions.

"It's much easier to follow — I don't feel like I'll lose my place."
— Nurse feedback from UserTesting.com session

Reflection

- **Translate real-world urgency into clear, reliable flows.**
- **Balance compliance and usability through documentation rigor.**
- **Use UserTesting.com feedback loops to validate clarity before code.**

The project strengthened my skill in regulated UX, blending design empathy with technical precision—an essential combination for industrial healthcare innovation.

Design Log for updates

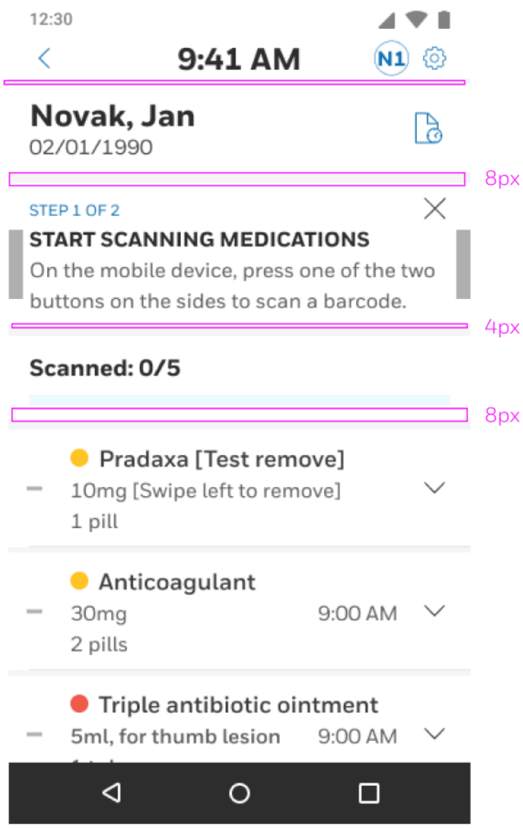
Iterative refinements were tracked through a design log:
• Button heights standardized to 48px for glove compatibility.
• Icons recolored to blue for improved visibility.
• Confirmation buttons relabeled (“End Session” vs. “Submit”) for clarity.
• In-app help consolidated under a new Profile screen, including onboarding replays and admin contacts.
• A success animation was added post-administration to reinforce completion and provide positive feedback.

March 28, 2023

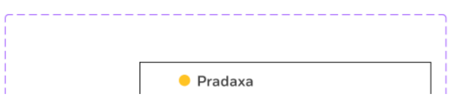
SCAN SCREEN

- 3/28/2023

Removed the gap between the header and changed spacing between. Changed spacing to 8px between sections and 4px between subsections.



Feb 22, 2023



Prototype

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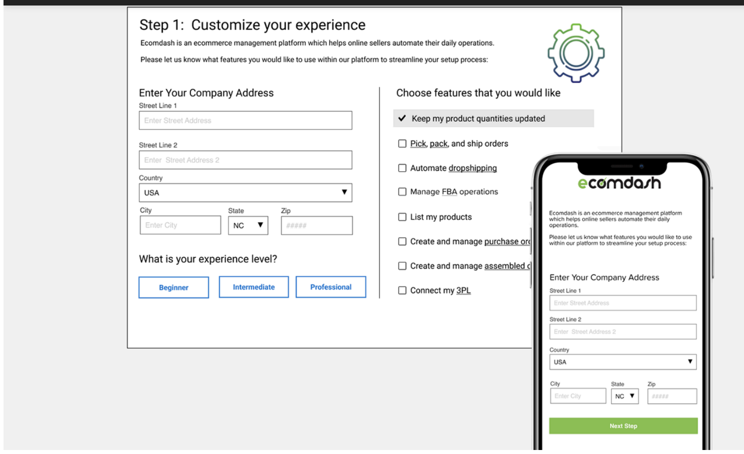

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